MTH321

Specimen Exam Paper

Engineering Mathematics II

Time allowed: 2 hours

INSTRUCTIONS TO STUDENTS:

1. This examination contains **FOUR (4)** questions and comprises **THREE (3)** printed pages (including cover page).

2. You must answer **ALL** questions.

3. All answers must be written in the answer book. Marks will only be awarded if FULL working is shown.

4. This is **an Open-book** examination.

At the end of the examination
Please ensure that you have written your examination number on each answer book used.

Failure to do so will mean that your work cannot be identified.

If you have used more than one answer book, please tie them together with the string provided.

THE UNIVERSITY RESERVES THE RIGHT NOT TO MARK YOUR SCRIPT IF YOU FAIL TO FOLLOW THESE INSTRUCTIONS.

Answer all questions. (Total 100 marks)

Question 1

A mass-spring-damper system is described by the following second-order differential equation:

$$m\frac{d^2x}{dt^2} + c\frac{dx}{dt} + kx = F(t)$$

where: m is the mass, c is the damping coefficient, k is the spring constant, $x(t)$ is the displacement of the mass as a function of time and $F(t)$ is the external force applied to the mass. The initial conditions are

$$x(0) = 0, x'(0) = 0$$

and

$$m = 1kg, c = 2Nsm^{-1}, k = 5Nm^{-1}, F(t) = 10\sin t$$

1 (a) Use Laplace transform, solve for $x(t)$

(18 marks)

1 (b) What happens to the transient response of the system?

(2 marks)

Question 2

Given the Laplace equation problem:

$$u_{xx} + u_{yy} = u_{tt}, \qquad 0 < x < 1, 0 < y < 2$$
$$u(0, y) = 0, u(1, y) = 1, \quad 0 < y < 2$$
$$u(x, 0) = 0, u(x, 2) = g(x), \quad 0 < x < 1$$

where

$$g(x) = \begin{cases} 2x, & 0 \le x \le \frac{1}{2} \\ 2(1-x), & 1/2 < x \le 1 \end{cases}$$

(a) 3 Are the boundary conditions homogeneous? If not, split the given problem into two Laplace equation problems.

(5 marks)

(b) 3 Hence solve the given problems.

(20 marks)

1a) $x(0) = 0, x'(0) = 0$

$m = 1kg, c = 2Nsm^{-1}, k = 5Nm^{-1}, F(t) = 10\sin t$

Sub into ODE $\rightarrow x'' + 2x' + 5x = 10\sin t$.

$f(t)$	$F(s)$
$g'(t)$	$sG(s) - g(0)$
$g''(t)$	$s^2G(s) - sg(0) - g'(0)$
$g^{(n)}(t)$	$s^nG(s) - \sum_{i=1}^{n} s^{n-i}g^{(i-1)}(0)$

$\sin(\omega t)$	$\frac{\omega}{s^2+\omega^2}$

$\mathcal{L}\{ODE\} \rightarrow s^2X - s(0) - 0 + 2[sX - 0] + 5X = \frac{10}{s^2+1}$

$X(\underbrace{s^2+2s+5}_{\text{cannot factorize}}) = \frac{10}{s^2+1}$

$X = \frac{10}{(s^2+1)(s^2+2s+5)} = \frac{As+B}{s^2+1} + \frac{Cs+D}{s^2+2s+5}$

$10 = (As+B)(s^2+2s+5) + (Cs+D)(s^2+1)$

$= s^3(A+C) + s^2(B+2A+D) + s(5A+2B+C) + 5B+D$

Comparing coeffs:

s^3: $0 = A+C \longrightarrow A = -C$

s^2: $0 = B+2A+D \rightarrow 0 = -5C+2B+C = 2B-4C.$

$\Rightarrow B = 2C$

s: $0 = 5A+2B+C \rightarrow 0 = 2C+2(-C)+D = D \rightarrow D = 0.$

const: $10 = 5B+D \rightarrow 10 = 5B+0 \rightarrow B = 2$

$C = 1$

$A = -1$

$\Rightarrow X(s) = \frac{-s+2}{s^2+1} + \frac{s}{s^2+2s+5} \rightarrow$ complete square $\rightarrow \underline{s^2+2s+1}-1+5$

$= \frac{-s}{s^2+1} + \frac{2}{s^2+1} + \frac{s+1-1}{(s+1)^2+4}$

$= \frac{-s}{s^2+1} + \frac{2}{s^2+1} + \frac{s+1}{(s+1)^2+4} - \frac{1(2)}{(s+1)^2+4}\left(\frac{1}{2}\right)$

$g(t)e^{at}$	$G(s-a)$

$\therefore x(t) = \mathcal{L}^{-1}\{X(s)\} = -\cos t + 2\sin t + \cos(2t)\cdot e^{-t} - \frac{1}{2}\sin(2t)\cdot e^{-t}$ //

b) Transient response is $\cos(2t)\cdot e^{-t} - \frac{1}{2}\sin(2t)\cdot e^{-t}$, which decays to zero as $t \rightarrow \infty$.

Question 3

(a) Given

2

$$f(x) = \begin{cases} 1 - x^2, -1 \le x \le 1 \\ 0 \qquad , otherwise \end{cases}$$

Plot its graph and find its Fourier transform using complex exponential formula.

(15 marks)

(b) Given a triangular wave function:

2

$$g(x) = |x|, -1 \le x \le 1$$
$$g(x) = g(x+2)$$

Plot its graph and find its Fourier series.

(15 marks)

Question 4

(a) Apply chain rule to find dy/dx and d^2y/dx^2 in terms of dy/dz and d^2y/dz^2 given that $z = \ln x$.

4

(10 marks)

(b) Show that the equation $y''x^2 + 2y'x - y = 0, x > 0$ can be converted into a linear equation with constant coefficients by substituting $z = \ln x$.

4

(5 marks)

(c) Hence determine its general solution.

4

(10 marks)

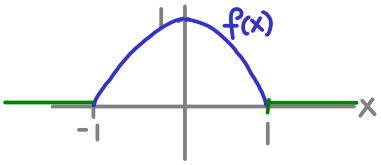
----- END OF PAPER -----

(a) Given

$$f(x) = \begin{cases} 1 - x^2, & -1 \le x \le 1 \\ 0 & , otherwise \end{cases}$$

Plot its graph and find its Fourier transform using complex exponential formula.

(15 marks)



Fourier transform

$$F\{f(t)\} = \int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt$$

$$\hat{f}(\omega) = F\{f(x)\} = \int_{-1}^{1} (1-x^2) e^{-i\omega x} dx$$

$$= \left\{ e^{-i\omega x} \left[\frac{x^2-1}{i\omega} - \frac{2x}{\omega^2} - \frac{2}{i\omega^3} \right] \right\} \Big|_{-1}^{1}$$

	u	dv
+	$1-x^2$	$e^{-i\omega x}$
−	$-2x$	$\frac{-1}{i\omega} e^{-i\omega x}$
+	-2	$\frac{-1}{\omega^2} e^{-i\omega x}$
	0	$\frac{1}{i\omega^3} e^{-i\omega x}$

$$\frac{-1}{i\omega}\left(\frac{-1}{i\omega}\right) e^{-i\omega x} = \frac{1}{i^2\omega} e^{-i\omega x} = -\frac{1}{\omega} e^{-i\omega x}$$

$$= e^{-i\omega}\left[0 - \frac{2}{\omega^2} - \frac{2}{i\omega^3}\right] - e^{i\omega}\left[0 + \frac{2}{\omega^2} - \frac{2}{i\omega^3}\right]$$

$$= \frac{2}{\omega^2}\left(-e^{-i\omega} - e^{i\omega}\right) + \frac{2}{i\omega^3}\left(-e^{-i\omega} + e^{i\omega}\right)$$

Since $\cos(\omega_n x) = \dfrac{e^{i\omega_n x} + e^{-i\omega_n x}}{2}$, $\sin(\omega_n x) = \dfrac{e^{i\omega_n x} - e^{-i\omega_n x}}{2i} = -i\dfrac{e^{i\omega_n x} - e^{-i\omega_n x}}{2}$

$$= \frac{-2(2)}{\omega^2}\underbrace{\left(\frac{e^{-i\omega} + e^{i\omega}}{2}\right)} + \frac{2(2)}{\omega^3}\underbrace{\left(\frac{e^{i\omega} - e^{-i\omega}}{2i}\right)}$$

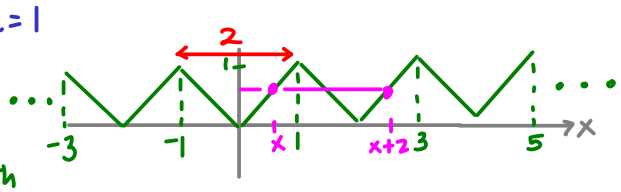
$$= -\frac{4}{\omega^2}\cos\omega + \frac{4}{\omega^3}\sin\omega \quad //$$

(b) Given a triangular wave function:

$L = 1$

$$\left.\begin{aligned} g(x) &= |x|, -1 \le x \le 1 \\ g(x) &= g(x+2) \end{aligned}\right\}$$

periodic with period 2.



Plot its graph and find its Fourier series.

(15 marks)

Since $g(x)$ is even, so $b_n = 0$.

$$a_0 = \frac{2}{1}\int_0^1 x\,dx = \cdots = 1.$$

$$a_n = \frac{2}{1}\int_0^1 x\cos(n\pi x)\,dx$$

$$= \frac{2}{1}\left[\frac{x}{n\pi}\sin(n\pi x) + \frac{1}{n^2\pi^2}\cos(n\pi x)\right]\Big|_0^1$$

$= 0$ at limits

	u	dv
+	x	$\cos(n\pi x)$
−	1	$\frac{1}{n\pi}\sin(n\pi x)$
	0	$\frac{-1}{n^2\pi^2}\cos(n\pi x)$

$$= \frac{2}{n^2\pi^2}\left[\cos(n\pi) - \cos 0\right]$$

$$= \frac{2}{n^2\pi^2}\left[(-1)^n - 1\right].$$

$$\therefore\ g(x) = \frac{1}{2} + \sum_{n=1}^{\infty} \frac{2}{n^2\pi^2}\left[(-1)^n - 1\right]\cos(n\pi x) \quad //$$

Similarly, for an **even function** f(x), we can define a Fourier cosine series given by:

All $b_n = 0$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{L}\right),$$

$$\text{where } a_0 = \frac{2}{L}\int_0^L f(x)dx, \quad a_n = \frac{2}{L}\int_0^L f(x)\cos\left(\frac{n\pi x}{L}\right)dx$$

Given the Laplace equation problem:

$$u_{xx} + u_{yy} = \cancel{u_{tt}}, \quad 0 < x < 1, 0 < y < 2$$

=0 at steady-state.

$$u(0,y) = 0, u(1,y) = 1, \quad 0 < y < 2$$

$L_2 = 2$

$$u(x,0) = 0, u(x,2) = g(x), \quad 0 < x < 1$$

$L_1 = 1$

Non-homo. BCs in both x & y.

where

$$g(x) = \begin{cases} 2x, & 0 \le x \le \frac{1}{2} \\ 2(1-x), & 1/2 < x \le 1 \end{cases}$$

a) Since the BCs are non-homo, we need to split up the sol$^{\underline{n}}$ to:

$$u(x,y) = u_1(x,y) + u_2(x,y).$$

where $u_1(x,y)$ & $u_2(x,y)$ are solved from:

$$u_{1xx} + u_{1yy} = 0,$$

homo BCs $\rightarrow$ $u_1(0,y) = 0, \; u_1(1,y) = 0,$
$u_1(x,0) = 0, \; u_1(x,2) = g(x).$

and

$$u_{2xx} + u_{2yy} = 0,$$

$u_2(0,y) = 0, \; u_2(1,y) = 1,$

homo BCs $\rightarrow$ $u_2(x,0) = 0, \; u_2(x,2) = 0.$

b) Sol$^{\underline{n}}$ is $u(x,y) = u_1(x,y) + u_2(x,y).$

For $u_1(x,y)$:

$$u_{1xx} + u_{1yy} = 0,$$

homo BCs $\rightarrow$ $u_1(0,y) = 0, \; u_1(1,y) = 0,$
$u_1(x,0) = 0, \; u_1(x,2) = g(x).$

$f(x) = 0$

homogeneous boundary conditions in x is:

$$u(x,y) = \sum_{n=1}^{\infty} \sin(\omega_n x)\left[a_n \cosh(\omega_n y) + b_n \sinh(\omega_n y)\right], \quad \omega_n = \frac{n\pi}{L_1}$$

where:

$$a_n = \frac{2}{L_1}\int_0^{L_1} f(x)\sin\left(\frac{n\pi x}{L_1}\right)dx$$

$$b_n = \frac{1}{\sinh(\omega_n L_2)}\left[\underbrace{\frac{2}{L_1}\int_0^{L_1} g(x)\sin\left(\frac{n\pi x}{L_1}\right)dx}_{C_n} - a_n\cosh(\omega_n L_2)\right]$$

$$g(x) = \begin{cases} 2x, & 0 \le x \le \frac{1}{2} \\ 2(1-x), & 1/2 < x \le 1 \end{cases}$$

Since $f(x) = 0$, $a_n = 0$.

$$C_n = \frac{4\cancel{2}}{1}\left[\int_0^{1/2} \cancel{2}x\sin(n\pi x)\,dx + \int_{1/2}^{1} \cancel{2}(1-x)\sin(n\pi x)\,dx\right]$$

	u	dv
+	x	$\sin(n\pi x)$
−	1	$-\frac{1}{n\pi}\cos(n\pi x)$
	0	$-\frac{1}{n^2\pi^2}\sin(n\pi x)$

	u	dv
+	$1-x$	$\sin(n\pi x)$
−	-1	$-\frac{1}{n\pi}\cos(n\pi x)$
	0	$-\frac{1}{n^2\pi^2}\sin(n\pi x)$

$$= 4\left\{\left[\frac{-x}{n\pi}\cos(n\pi x) + \frac{1}{n^2\pi^2}\sin(n\pi x)\right]\Bigg|_0^{1/2} + \left[\frac{x-1}{n\pi}\cos(n\pi x) - \frac{1}{n^2\pi^2}\sin(n\pi x)\right]\Bigg|_{\frac{1}{2}}^{1}\right\}$$

$$= 4\left\{\left[\cancel{\frac{-1}{2n\pi}\cos\left(\frac{n\pi}{2}\right)} + \frac{1}{n^2\pi^2}\sin\left(\frac{n\pi}{2}\right) - 0\right] + \left[0 - \cancel{\left(\frac{-1}{2n\pi}\right)\cos\left(\frac{n\pi}{2}\right)} + \frac{1}{n^2\pi^2}\sin\left(\frac{n\pi}{2}\right)\right]\right\}$$

$$= \frac{8}{n^2\pi^2}\sin\left(\frac{n\pi}{2}\right).$$

$$\Rightarrow b_n = \frac{1}{\sinh(2n\pi)}\frac{8}{n^2\pi^2}\sin\left(\frac{n\pi}{2}\right) = \frac{8}{n^2\pi^2\sinh(2n\pi)}\sin\left(\frac{n\pi}{2}\right)$$

$$\therefore \; u_1(x,y) = \sum_{n=1}^{\infty} \frac{8}{n^2\pi^2\sinh(2n\pi)}\sin\left(\frac{n\pi}{2}\right)\sin(n\pi x)\cdot\sinh(n\pi y)$$

homogeneous boundary conditions in x is:

$$u(x,y) = \sum_{n=1}^{\infty} \sin(\omega_n x)\left[\cancel{a_n}^{\,0}\cosh(\omega_n y) + b_n \sinh(\omega_n y)\right], \quad \omega_n = \frac{n\pi}{L_1}$$

where:

$$a_n = \frac{2}{L_1}\int_0^{L_1} f(x)\sin\left(\frac{n\pi x}{L_1}\right)dx$$

$$b_n = \frac{1}{\sinh(\omega_n L_2)}\left[\frac{2}{L_1}\int_0^{L_1} g(x)\sin\left(\frac{n\pi x}{L_1}\right)dx - a_n\cosh(\omega_n L_2)\right]$$

Then we can expect the complete solution would be:

$$u(x,y) = \sum_{n=1}^{\infty} \sin(\omega_n y)\left[a_n \cosh(\omega_n x) + b_n \sinh(\omega_n x)\right], \quad \omega_n = \frac{n\pi}{L_2}$$

where:

$$a_n = \frac{2}{L_2}\int_0^{L_2} f(y)\sin\left(\frac{n\pi y}{L_2}\right)dy$$

$$b_n = \frac{1}{\sinh(\omega_n L_1)}\left[\underbrace{\frac{2}{L_2}\int_0^{L_2} g(y)\sin\left(\frac{n\pi y}{L_2}\right)dy}_{C_n} - a_n\cosh(\omega_n L_1)\right]$$

For $u_2(x,y)$:

$f(y)=0$ $\quad u_{2xx} + u_{2yy} = 0,$ $\quad g(y)=1$

$u_2(0,y) = 0, \quad u_2(1,y) = 1,$

homo BCs $\rightarrow$ $u_2(x,0) = 0, \quad u_2(x,2) = 0.$

Since $f(y)=0$, $a_n = 0$.

$$C_n = \frac{2}{2}\int_0^2 \sin\left(\frac{n\pi y}{2}\right)dy = \frac{-2}{n\pi}\cos\left(\frac{n\pi y}{2}\right)\Big|_0^2 = -\frac{2}{n\pi}\left[\cos(n\pi) - \cos(0)\right] = \frac{2}{n\pi}\left[1-(-1)^n\right]$$

$$\Rightarrow b_n = \frac{1}{\sinh\left(\frac{n\pi}{2}\right)}\frac{2}{n\pi}\left[1-(-1)^n\right] = \frac{2\left[1-(-1)^n\right]}{n\pi\sinh\left(\frac{n\pi}{2}\right)}.$$

$$\therefore u_2(x,y) = \sum_{n=1}^{\infty}\frac{2\left[1-(-1)^n\right]}{n\pi\sinh\left(\frac{n\pi}{2}\right)}\sin\left(\frac{n\pi y}{2}\right)\sinh\left(\frac{n\pi x}{2}\right).$$

Then we can expect the complete solution would be:

$$u(x,y) = \sum_{n=1}^{\infty} \sin(\omega_n y)\left[\cancel{a_n}^{\,0} \cosh(\omega_n x) + b_n \sinh(\omega_n x)\right], \quad \omega_n = \frac{n\pi}{L_2}$$

where:

$$a_n = \frac{2}{L_2}\int_0^{L_2} f(y)\sin\left(\frac{n\pi y}{L_2}\right)dy$$

$$b_n = \frac{1}{\sinh(\omega_n L_1)}\left[\frac{2}{L_2}\int_0^{L_2} g(y)\sin\left(\frac{n\pi y}{L_2}\right)dy - a_n\cosh(\omega_n L_1)\right]$$

Complete sol$^{\underline{n}}$ is:

$$u(x,y) = u_1(x,y) + u_2(x,y)$$

Question 4

(a) 4 Apply chain rule to find dy/dx and d^2y/dx^2 in terms of dy/dz and d^2y/dz^2 given that $\underbrace{z = \ln x}_{z(x)}$. (10 marks)

$z(x) \longrightarrow \frac{dz}{dx} = \frac{1}{x}.$

Implied that $y(x) = y(z(x))$

$\frac{d}{dx}\downarrow \qquad \downarrow\frac{d}{dx}$ (chain-rule)

$$\frac{dy}{dx} = \frac{dy}{dz}\cdot\frac{dz}{dx} = \frac{dy}{dz}\cdot\frac{1}{x}$$

$\frac{dy}{dz}$ is a function of $z(x)$.

$\frac{dy}{dz}(z(x))$

$\frac{d}{dx}\downarrow \qquad \downarrow\frac{d}{dx}$ (chain-rule)

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left\{\frac{dy}{dz}\cdot\frac{1}{x}\right\} = \frac{1}{x}\frac{d}{dx}\left\{\frac{dy}{dz}\right\} + \frac{dy}{dz}\cdot\frac{d}{dx}\left\{\frac{1}{x}\right\}$$

product rule $\qquad$ chain rule

$$= \frac{1}{x}\left\{\frac{d}{dz}\left(\frac{dy}{dz}\right)\cdot\frac{dz}{dx}\right\} + \frac{dy}{dz}\left(-\frac{1}{x^2}\right)$$

$$= \frac{1}{x}\left\{\frac{d^2y}{dz^2}\cdot\frac{1}{x}\right\} - \frac{1}{x^2}\frac{dy}{dz} = \frac{1}{x^2}\left(\frac{d^2y}{dz^2} - \frac{dy}{dz}\right)$$

(b) 4 Show that the equation $y''x^2 + 2y'x - y = 0, x > 0$ can be converted into a linear equation with constant coefficients by substituting $z = \ln x$.

(5 marks)

(c) 4 Hence determine its general solution.

(10 marks)

b) $y''x^2 + 2y'x - y = 0 \leftarrow$ ODE.

Sub $y' = \frac{dy}{dz} \cdot \frac{1}{x}$ & $y'' = \frac{1}{x^2}\left(\frac{d^2y}{dz^2} - \frac{dy}{dz}\right)$ into ODE,

$$\cancel{\frac{1}{x^2}}\left(\frac{d^2y}{dz^2} - \frac{dy}{dz}\right)\cancel{x^2} + 2\frac{dy}{dz} \cdot \cancel{\frac{1}{x}}\cancel{x} - y = 0$$

$$\Rightarrow \frac{d^2y}{dz^2} + \frac{dy}{dz} - y = 0 \quad // \leftarrow \text{homo 2nd-order ODE.}$$

c) Aux eqn is $\lambda^2 + \lambda - 1 = 0$

$$\lambda_{1,2} = \frac{-1 \pm \sqrt{1 - 4(1)(-1)}}{2(1)} = -\frac{1}{2} \pm \frac{\sqrt{5}}{2}$$

$$\Rightarrow \lambda_1 = -\frac{1}{2} + \frac{\sqrt{5}}{2}, \quad \lambda_2 = -\frac{1}{2} - \frac{\sqrt{5}}{2}.$$

Since λs are real & distinct,

$$y(z) = Ae^{\lambda_1 z} + Be^{\lambda_2 z}.$$

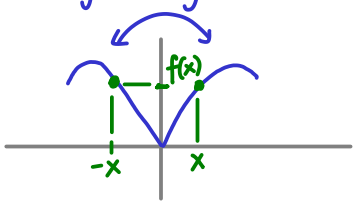
Since $z = \ln x$,

$$y(x) = Ae^{\lambda_1 \ln x} + Be^{\lambda_2 \ln x}$$

$$= Ae^{\ln x^{\lambda_1}} + Be^{\ln x^{\lambda_2}} = Ax^{\lambda_1} + Bx^{\lambda_2} \quad //$$

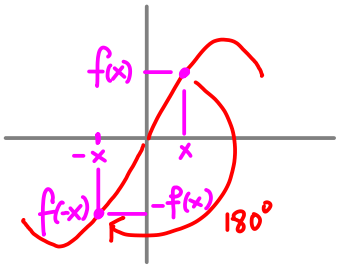
Even Function ($b_n = 0$)

Mirror symmetry abt y-axis



$f(-x) = f(x)$

Odd Function ($a_0 = 0$, $a_n = 0$)



$f(-x) = -f(x)$.

$E \times O = O$. $(x^2 \times x = x^3)$

$E \times E = E$. $(x^2 \times x^4 = x^6)$

$O \times O = E$. $(x \times x^3 = x^4)$